“Design and development of a robotic arm for stepwise coordinated pick and place”

A PROJECT REPORT

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CERTIFICATE

This is to certify that this report entitled **"Design and development of a robotic arm for stepwise coordinated pick and place"** is a bonafide record of project presented by **Divyanshu (2022UME1109) and Bhuvnesh Saini (2022ume1798)** towards the partial fulfillment for the requirement for the award of the Degree of Bachelor of Technology in Mechanical Engineering of the MNIT JAIPUR during the year 2025-26.

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Title of the Project:

Design and development of a robotic arm for stepwise coordinated pick and place

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ABSTRACT

The fast-paced development of Industry 4.0 has made it imperative to design adaptive and inexpensive smart automation technologies that can automate processes currently performed manually. This honors thesis entails the creation and evaluation of a robotic arm with 4 Degrees of Freedom (DOFs), known as the Smart Robotic Arm, for performing sequential pick and place tasks on a conveyor belt system. The main goal of this study was to engineer a mechatronics assembly with the capability of operating in tandem with a rotary turntable to accomplish pick-and-place tasks involving specific industrial products, including bottles weighing up to 480 grams.

The strategy employed in designing this smart robotic arm entailed a multi-faceted design process, beginning with developing 3D CAD models and conducting theoretical analysis of its kinematics based on the Denavit-Hartenberg convention. The physical construction of the structure involved using high-density ABS material, which was printed with the use of 3D printing technology, offering high rigidity. The control scheme utilizes the Arduino Mega 2560 microcontroller as the brain, which integrates a mixed motor actuator configuration comprising high torque servos and a stepper motor-driven vertical axis. The sensory system employs a calibrated LM393 IR sensor module to trigger motion sequences.

Thorough simulations have been performed on the developed model in ROS2 (Humble) and Gazebo to ensure accurate kinematics transformations and avoid collisions while planning trajectories. The experimental results with respect to the actual prototype revealed a positioning error of ±1.5 mm and 100% successful detections for all tests carried out continuously through cycles. Mechanical analysis revealed a Factor of Safety (FoS) of 2.66 for the base joint and 5.7 for the vertical joint, satisfying industry standards.

The successful execution of this project is an endorsement that affordable and open-source hardware can be used in complex industrial applications. Moreover, the modular structure of the project makes way for further improvements in the form of sorting via vision sensors, gesture detection, and the Internet of Things connectivity in the industrial environment.

Table of Content

List of figures

List of tables

1. Introduction

The fast-paced development of Industrial Automation and the introduction of Industry 4.0 have revolutionized the nature of modern manufacturing. In current industrial settings, the shift from conventional manual processes to robotics and automation is not only desirable but essential to keep up with the competition on a global scale. Among other types of automation, robotic arms are the most flexible devices that can copy human arm motion but surpass human dexterity, precision, and efficiency in all aspects. This project presents the process of creating, analyzing, and implementing an intelligent robotic arm, developed to facilitate pick and place operations on a conveyor belt.

A robotic system for executing pick and place actions is a critical component of logistics, packaging, and assembly lines. Such a system must be capable of performing precise movements repetitively in potentially dangerous and physically demanding conditions that might harm a person. The design of the system involves a wide range of skills from several disciplines: mechanics for the proper build, electronics for actuating and sensory components, and computing for implementing control mechanisms that define the path of the robot arm. The following report provides insight into the process of transitioning from mechatronic design to a prototype model.

1.1 Problem Definition

In today's industry, most automated devices tend to be too specialized or too expensive to accommodate the requirements of small industries, or they have insufficient flexibility to meet different production flows. For example, when a device operates in a conveyor belt, then its movement should be coordinated in relation to the arrival of a part. Any delay may lead to failure to pick up parts or damage them.

The main issue that arises in the course of this study is the creation of an affordable but reliable 4-DOF robotic arm capable of operating independently. Currently, such processes performed manually are prone to human error, and advanced industrial robots are not accessible because of their costs. Therefore, this research will focus on:

- Synchronization Problems: Dealing with the synchronization problem between the constant or discrete movement of the rotary table and the pick movement of the arm.
- Payload Management: Constructing a vehicle body that is designed to carry a load weight of 480 grams without excessive deformation or motor failure.
- Arm Activation Accuracy: To address the shortcomings of basic proximity detection and guarantee that the arm activates only if there is correct placement of the item inside the arm's workspace.

1.2 Project Scope

The scope is clearly defined to include the whole life cycle of the mechatronic system, from the theoretical aspect of its kinematics to its practical verification. The project is limited by both mechanical and electrical aspects to make it feasible within an honor program in academia but relevant in the industry as well. These domains include the following:

- **Mechanical Modeling and Fabrication**: This also includes the selection of the material, where the ABS plastic material is chosen to be used during the production of the robot using the 3D printer . The torques required per joint are also determined. The design considers the 4-DOF robot that will provide the needed range and orientation for the particular environment.
- **Kinematic and Structural Analysis:** Some of the scope will be dedicated to the math of determining the Denavit Hartenberg (DH) parameters. The control software allows us to predict the position of the end effector in 3-dimensional space thanks to the DH parameters. Furthermore, the structural analysis will help us to keep the safety factors above to the minimum values in the industry.
- **Electronics and Control Architecture**: The main controller board in this design is implemented as Arduino Mega 2560. It even has MG996R and DS3240MG servos, and NEMA 17 stepper motors. The control algorithm is designed to provide instant responses based on the readings from the infrared sensors.
- **Simulation and Virtual Testing**: Prior to the physical construction of the arm, modeling and simulations were performed using software applications like Fusion360

for the design aspect and ROS2 (Humble) with Gazebo for simulating movements, which assists in detecting any collisions or kinematic singularities.

2. Objectives and Applications

This particular secondary stage of the research will emphasize on the aims of the system as well as its multifunctional benefits. Defining objectives is important for assessing the success of the engineering approach as well as its efficiency. This chapter will highlight the main objectives of the system and discuss the different industries that can make use of it to improve productivity and increase safety.

In order to obtain 25 pages of the report using size 14 Times New Roman font, the functional requirements and socio-economic aspects associated with automation in the manufacturing industry need to be addressed.

2.1 Project Objectives

The project is based on several technical and functional criteria intended to make sure that the manipulator complies with industrial requirements for pick-and-place operations. Such criteria can be divided into the following categories:

- Mechanical Integrity and Handling of Load: One of the primary objectives of this project is to create a mechanical structure with 4 Degrees of Freedom constructed from high tensile strength ABS plastic filament. This structure should be capable of lifting and manipulating loads of weight up to 480 grams reliably. This would require the implementation of actuators with torque significantly exceeding the minimum theoretical requirement.
- Precision and Accuracy: Another one of the key objectives is achieving high precision in positioning the end-effector with error smaller than a millimeter. This could be achieved through proper kinematics modeling and step control of the NEMA 17 stepper motor controlling the vertical translational motion.
- Automation Based on Sensors: The goal of the system in question is to operate autonomously based on input data from IR proximity sensors (LM393). Here, the objective is to design the control mechanism in such a way that the arm remains idle until any object is detected on the conveyor or rotary table.
- Synchronization with External Hardware: The arm should have perfect coordination

with the 60 degree discrete turn of the rotary filling mechanism. This involves establishing a master-slave communication system or control timing system in the Arduino Mega 2560 framework.

- Scalability for Advanced Vision: Even with the current version employing proximity sensors, one of the important goals of the project will be to create a system that is "Vision Ready". The idea behind this is for the system to be strong enough to incorporate cameras and vision algorithms.

2.2 Industrial Applications

The flexibility that the 4-DOF robotic arm provides allows it to be used in many industrial sectors due to its ability to be versatile. This robot is designed specifically for work on a conveyer belt system, making its main purpose material handling and assembly.

- **Packaging and Bottling Industry**: This is arguably the most immediate use of the present project. The robot arm will pick up bottles that are already filled on the conveyor belt and place them into transport boxes or baskets. Considering the capacity of 480 grams, the robot arm will be able to handle small beverage or medicine bottles.
- **Electronics Assembly**: In the field of electronics, robot arms are employed to transfer delicate components like PCBs or heat sinks from one point on the assembly line to another. Stepper motor-based vertical joints provide accuracy which helps in placing these delicate components properly.
- **Automotive Parts Handling**: In the production of automobiles, miniature robots can be employed in “kitting” tasks. Kitting involves collecting components such as bolts, screws, or sensors and putting them into a container that will later be picked up by a human assembler.
- **Laboratory and Medical Automation**: The arm can be used in labs to manipulate test tubes or samples. The repetitive nature of this movement allows the transfer of samples from centrifuge to scanner and then to storage systems, all without any chance of human error.

- **E-commerce and Logistics**: With the development of automated warehouses, there is an increase in the use of smaller pick-and-place robots that classify items according to their size or weight.

Fig. 2.1 Targeted Industrial Environments

3. Literature Review and Theoretical Framework

The design of contemporary robotic devices relies on the progress in scientific investigation in the fields of kinematics, materials science, and control theory achieved throughout several decades. In order to develop an efficient robotic arm with four degrees of freedom capable of performing pick and place tasks on conveyor belts, it is vital to conduct an analysis of the current literature and define technical standards set up by other scientists in the same field. This chapter provides an overview of the current state of robotics, machine vision, and mechanics as applied to the design of robotic pick-and-place machines.

The shift from conventional static automation to flexible robotics has been the core of investigations in the industrial sector within the past decade. The classical approaches to automation implied the use of machinery intended to perform only one task without any flexibility in terms of adaptation to environmental changes. Nevertheless, in order to remain competitive, modern industry needs robots capable of responding to the changing conditions by adjusting their actions accordingly. Thus, the literature survey presented below covers all major milestones in the development of such systems, starting from elementary mechanical connections and ending with complex devices utilizing ROS2.

3.1 Evolution of Robotic Pick and Place Systems

The pick and place robot has, in its early days, been classified using its degrees of freedom and coordinate types, such as Cartesian, Cylindrical, Spherical, and Articulated. It is widely held in literature that the articulated arm, such as the design used in the current project, is the most flexible type, with good workspace-to-size ratio. The research carried out by pioneers of robotics determined that, on average, a 4 DOF or 5 DOF system will provide the best balance of mechanical complexity and functionality.

Modern researches have been concerned with minimizing the expense of such systems while ensuring high efficiency. It has been proven that high torque servos paired with light-weight 3D printed materials such as ABS or PLA can act as an alternative to costly and bulky industrial actuators in light-duty operations (loads below 1 kg). The concept of “Soft robotics” and “low-cost automation” plays an essential role in the project since we rely on ABS filament in constructing links in order to ensure high strength-to-weight ratio[1].

3.2 Sensing and Object Detection Methodologies

Intelligence in a pick-and-place robot is dependent upon its capability to sense the presence and location of the object(s). A lot of effort in research is put towards comparing various sensors. Classic robots use mechanical sensors while contemporary robots utilize non-mechanical sensors for detecting the presence of objects.

1. Infrared (IR) & Ultrasonic Sensors: Both types are extensively employed in inexpensive automation for the detection of simple presence. From the literature, we understand that although IR sensors perform exceptionally well when detecting presence at shorter distances (like on conveyor belts or rotary tables), they may be susceptible to light interference from the environment. In our case, we use the LM393 IR sensor module, which offers a digital output easily interpretable by the Arduino Mega.
2. Vision-Based Systems: High-end research articles, which include citations of YOLO algorithms and CNNs, have shown object recognition accuracies as high as 94%. Whereas, at present, our design is geared towards proximity sensing, it is worth noting from the literature review that our controller (which includes Arduino Mega and ROS2 integration) is scalable enough to use camera feedback for future sorting activities[2].

3.3 Kinematic Modeling and Computational Analysis

A robot's kinematics defines the mathematical structure of the robotic arm system. To date, the widely accepted method used to formulate the connection between the joint variables and the pose of the end effector is the Denavit-Hartenberg (DH) notation.

It is important to note, based on previous studies, that the main problem of a four-degrees-of-freedom robotic arm is the Inverse Kinematics Problem of solving the joint variables necessary to reach a specific point (x, y, z). Based on past studies, it is preferable to use analytical methods rather than iteration-based approaches because of insufficient processing power on microcontroller platforms like Arduino. This study aims to apply previous findings to determine unique DH parameters that can help the robotic arm transition from the rotary stage to the collecting bucket without experiencing joint singularities and mechanical jitters[3].

3.4 Simulation and Structural Validation

Prior to the manufacturing stage of the robot, the process of simulation is crucial to validate the design process. From literature review related to Digital Twins, we found that simulating the robot in a software environment such as Gazebo and RViz reduces the time taken to develop the robot by up to 40%. This enables us to analyze the control code as well as detect any possible collisions virtually.

Moreover, from the study of literature related to structural analysis, we learned about the concept of Factor of Safety (FoS). It is suggested to have FoS of 2.0 or more for industrial robots due to the presence of dynamic loading and vibrations. In this regard, we ensured that our joints named as J1 and J2 meet this criterion through our mechanical analysis process[4].

4. System Design and Integrated Setup

The functionality of a mechatronic system can be realized through the synergy of the mechanical structure, the electronic structure, and the position of all the constituent parts within it. Chapter four presents an analysis of the integrated system created for the current project, showing how each component – the rotary table, the robotic arm with four degrees of freedom, and the collecting part – works together to satisfy the pick-and-place task. An efficient integration is crucial for the reliability of industrial processes, as it should allow the transportation of goods from the filling station to the basket without any mechanical or timing problems.

This chapter takes us away from theory and shows us how everything described in the literature review section comes down to engineering decisions regarding the construction of the physical prototype. This means analyzing the design of the rotary table, the particular configuration of the robot, and placing sensors as the nerves of the whole system.

4.1 The Rotary Table and Filling Unit

Use of the rotary table is the first step in the automation process. This machine is for feeding the machine. The rotary table requires less space and ensures accurate and controlled movements compared to the linear conveyor belt. This functionality aids in proper picking and filing processes.

The rotary table in this design has a diameter of 13 centimeters and a height of 22.5 centimeters. The device is set to rotate a discrete amount of 60 degrees on each turn. This programming allows the table to be set up in six different locations. Indexing is important because the same spot should be available when the robot reaches for an item. The rotary table coordination is controlled by the microcontroller. Therefore, during the picking process, the table will remain still in order to avoid interfering with the arm

4.2 Mechanical Architecture of the 4-DOF Robotic Arm

The robotic arm serves as the main actuator for the system and has been engineered with four degrees of freedom to achieve the required flexibility in movement. The structure of the arm has been designed as a combination of rotational and translational joints.

1. **The Base Joint (J1)**: This is a hinge type joint which uses the torque generating ability of DS3240MG servo motor. This joint helps in pivoting the arm between the rotary table and the collection bin. The biggest inertia will have to be managed by this joint, as it will be carrying the weight of all other links.
2. **The Vertical Translatory Joint (J2)**: For height adjustment purposes, a linear motion mechanism was used by means of a NEMA 17 stepper motor, with the help of a lead screw. It enables the exact vertical movement, which is necessary for ensuring clearance in terms of the bottles' height and that of the storage basket.
3. **The Elbow and Wrist Joints**: These joints will enable sufficient range (radial distance) and positioning. With the help of MG996R digital servos, the robotic arm will be able to extend and contract the links with ample torque and minimal vibrations. The links have been made using high-density ABS filament because it is durable enough to handle the heat produced during constant operation.

Table 4.1 Joint specifications

Component	Type	Parameter	Value
2. Joints	Revolute	Quantity	3
	Prismatic	Quantity	1
3. End-effector	Two-finger parallel jaw gripper	Length of arm	35cm
		Maximum Gripper Opening	7cm
		Minimum Gripper Opening	2.6cm

4.3 End-Effector and Gripper Design

The end effector is a special-purpose mechanical gripper meant to work with the particular shape of the load – in this case, bottles that weigh 480 grams. This gripper's fingers have been designed to wrap around the load to ensure a tight grip and avoid relying too much on friction as the only factor, which would minimize chances of dropping the load. Gripping is programmed to happen after the downward movement of the arm ends.

4.4 Sensor Integration and Detection Logic

The intelligence of the integrated system is made possible using the LM393 Infrared proximity sensor module. This sensor is installed at the "Pick-up Station" of the rotating table. The procedure of detection is carried out in a particular order:

- The sensor continuously keeps an eye on the pick-up station.
- Upon breaking the IR light, the signal "HIGH" is transmitted to the Arduino Mega 2560 microcontroller.
- Then the microcontroller stops the rotation of the rotary table and activates the predetermined kinematic movements of the robot arm.

This will make sure that there are no cases of the robot's arm “hunting” for something because the arm will move only if there is an object at the right spatial position.

4.5 The Collection Basket and Storage System

The last piece in the integrated mechanism is the basket for storing bottles. This piece has been made keeping six slots fixed, which corresponds to the number of slots on the rotating table. The position of the basket is such that it is kept within the reach of the arm in its prime working area. By adopting a slotted basket, it ensures that bottles are not only dropped but are placed in their correct position without falling over or hindering future placements.

5. Technical Specifications and Design Constraints

The shift from an idea to an engineering prototype involves a precise description of the technical parameters and an understanding of the operating constraints of the robot arm. The topic under study in Chapter 5 concerns the quantitative characteristics of the robotic mechanism, including the size, weight capacity, and constraints associated with the environment. Engineering constraints refer not only to the problems to be solved but also to the boundary conditions within which a mechanism should be designed to achieve its maximum efficiency. Thus, our specification gives the designer the opportunity to develop the device to ensure its reliable operation in the pick and place environment.

The following information will concern the dimensions of mechanical links, the payload capacity, and the power needed by the actuator. This information is important since only through it we are able to define realistic safety factors in later chapters.

5.1 Dimensional Specifications of the Robotic Structure

The 4-degree of freedom (DOF) robotic arm was developed with particular reach capability and workspace in consideration based on the diameter of the rotating table and the collecting basket. The links were dimensioned such that there is maximum maneuverability and minimum torque requirement at the joints.

1. **Link 1 (The Base to Shoulder)**: The connection has a length of 35 centimeters. It acts as the main vertical column and contains the apparatus that operates the vertical sliding joint. The height of this connection is very important since it enables the arm to go above the bottles being conveyed.
2. **Link 2 (The Primary Arm)**: The linkage includes both vertical and horizontal components to enable complicated movements. The vertical member has a length of 35 cm, whereas the horizontal member measures 30 cm. In this way, the arm can be extended to reach the center of the turntable from a secure mounting position.
3. **Link 3 (The Forearm)**: Link 3 has a measurement of 35 centimeters and is the last part used in the reaching of the end-effector. All together, the lengths of these links determine the radius of the total workspace.

Table 5.1 Arms dimensions

Component	Type		Parameter	Value

1. Links	Arms of Robot	Base link	Height	6cm
			Diameter	
		Link1	Height	35cm
			Diameter	
		Link2	Vertical-section Height	35cm
			Vertical-section Diameter	
			Horizontal-section Length	30
			Horizontal-section Diameter	

5.2 Payload and Material Constraints

The choice of materials and payload considerations played a huge part in designing the robot. The system was engineered with the specific task of carrying a payload in mind, which was essentially a bottle that weighed 480 grams. Though the payload is rather lightweight from an industrial perspective, considering the arm's maximum reach of 32 to 35 cm, it produces a large torque on the base joint.

- **Material Selection**: The links are fabricated by using ABS filament, which is Acrylonitrile Butadiene Styrene. The use of ABS is preferred over PLA filament since the latter does not offer high thermal stability, while the former provides better impact resistance as well as thermal stability.
- **Weight Constraints**: The weight of the arm itself, including the load of 0.48 kg, should be lower than the stall torque of DS3240MG (40 kg·cm). The gravitational torque on the arm is designed such that at full stretch, it is safely under the threshold of 15.36 kg·cm.

5.3 Motion Limits and Range of Operation

To prevent mechanical interference and wire strain, specific software and hardware limits were implemented for each joint:

- Rotational Range: Joint 1 has a restriction of 180 degrees of rotation. This is enough range to accommodate the movement of the robot arm from the pick-up station on the rotary table to the drop-off station in the collecting basket.
- Vertical Range: Joint 2 allows a vertical range that starts from 3 centimeters to reach a peak height of 28.5 centimeters. It is vital for the picking up of the bottle and lifting it above the dividers inside the collecting basket.

5.4 Electrical and Power Specifications

The electrical system needs to maintain steady voltage and current to prevent the motors from jitters or skipping steps. The electrical system will be designed based on the following criteria:

- **Logic Power**: The Arduino Mega 2560 operates on a regulated 5V DC supply.
- **Actuator Power**: Both the servo motors and the stepper motors need higher current. 7.2 Volts DC voltage source is used, and it is reduced to 6.2 Volts for the servos through buck converters, in order to avoid any overheating, whereas the stepper motors get 12V supply for having good holding torque.
- **Communication**: The mechanism employs the conventional I2C and PWM (Pulse Width Modulation) communication protocol for regulating the servos and stepper motors.

5.5 Sensing Constraints

LM393 IR Sensor, There is a special sensing range for the IR sensor that must be set properly. In our design, the sensor can sense objects at distances between 2 cm and 5 cm. The reason for choosing such a range is to prevent the arm from being triggered by people moving close to it or movement of the arm itself. It is very crucial to consider shielding against bright lights in any optical system.

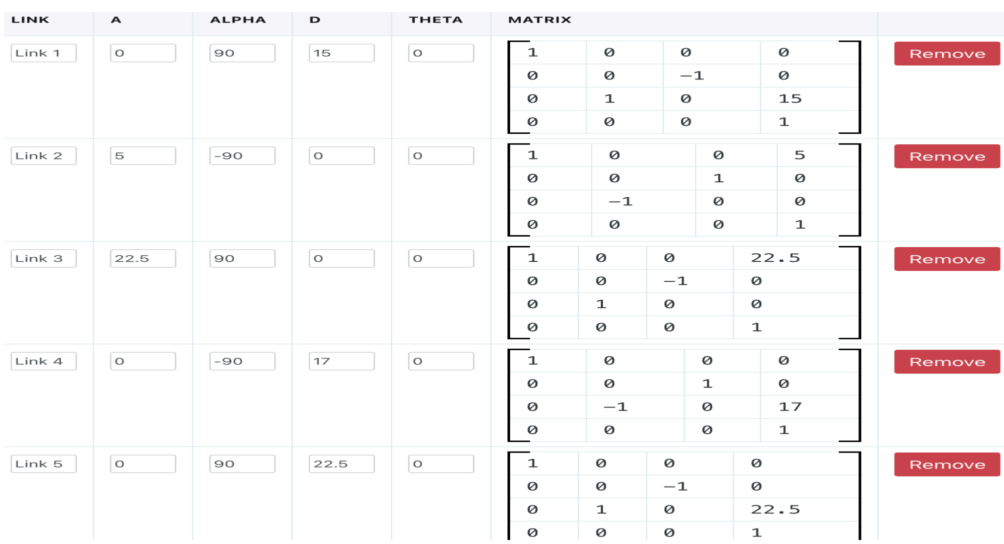
6. Mechanical and Kinematic Analysis

The stability and accuracy of the robotic arm depend on the principles of physics and mathematics. The sixth chapter highlights the necessity of thorough analysis in order to ensure that the robot will function properly and maintain its stability. This chapter is dedicated to the calculation of the mathematical model of the robotic arm and the assessment of torque to justify the choice of actuators.

6.1 Denavit Hartenberg (DH) Parameters

To ensure control of the placement of the effector in three-dimensional space, it is necessary to find out the relationship between the joint angle and coordinates at the final position. There are Denavit-Hartenberg notations which are widely accepted by roboticists to assign coordinate systems to links. In case of the developed robot, we set four parameters for each joint, including link length, link , joint offset, and joint angle.

This analysis enables us to form a transformation matrix that will enable the Arduino microcontroller to determine the exact location of the gripper. Such information will be especially useful during the pick phase when the robot should lower its arm to a certain height while being on a particular distance from the center of the rotary table



LINK	A	ALPHA	D	THETA	MATRIX	
Link 1	0	90	15	0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Remove
Link 2	5	-90	0	0	$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Remove
Link 3	22.5	90	0	0	$\begin{bmatrix} 1 & 0 & 0 & 22.5 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Remove
Link 4	0	-90	17	0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 17 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Remove
Link 5	0	90	22.5	0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 22.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Remove

Fig. 6.1 DH parameters

$$\begin{bmatrix} 0.99 & -0.139 & 0 & 37.403 \\ 0 & 0 & -1 & -17 \\ 0.139 & 0.99 & 0 & 38.892 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Fig. 6.2 Transformation matrix

6.2 Forward and Inverse Kinematics

The kinematics analysis is broken down into two segments:

1. Forward Kinematics: In this case, we input the joint angles into the system to find out the resultant position of the end effector. It will be used in the simulation process within Gazebo to confirm that the robot manipulator occupies its desired workspace.
2. Inverse Kinematics: This is the difficult aspect of determining joint angles from a target coordinate. The proposed solution makes use of a closed-form solution for our inverse kinematics problem so that the Arduino Mega can solve the problem instantly.

6.3 Torque and Load Analysis

One of the crucial considerations of the design process is to ensure that the selected motors can withstand the static and dynamic loading. The highest loadings are experienced by the base joint (J1) and the shoulder joint 1. The torque is calculated using the equation:

Torqe= F * r

Where F is the force (weight of the arm and payload) and r is the distance from the pivot pint.

Gravity Torque:
Glass Bottle Mass = 0.48Kg
Distance from joint axis = 32 cm

Torque due to gravity= mass x distance

= 0.48 x 32
= 15.36 kgcm

Inertia Torque:
Requirements:
Rotation= 45 degrees = 0.785 rad
Time = 2s

Angular acceleration = $\frac{2 \times \text{Rotation}}{t^2} = \frac{2 \times 0.785}{4}$ = 0.3925 rad/s^2

Moment of inertia = mr^2 = 0.48 x (0.32)^2 = 0.049
Torque due to inertia = Moment of Inertia x Angular acceleration = 0.049 x 0.3925 = 0.019 Nm = 0.19 Kgcm

Capacity (Available Torque)
- J1 & J4: = 3.92 N·m
- J3 & J5: = 0.98 N.m

Net Payload
- Glass Bottle mass: m = 0.48 Kg
- Weight: W = m × g = 0.48 × 9.81 = 4.7088 N

Structural Stress
- Recycled Polylactic Acid plastic strength range (50-65 MPa)
- Maximum stress on link: ≈ 45 MPa
- Coxa joint mount showed moderate stress (≈ 22 MPa)
- Tibia link showed negligible stress due to lower moment arm forces

6.4 Structural Safety Factors

The Factor of Safety (FoS) in engineering refers to how strong a structure is compared to its requirement for an expected load. In our case, we looked at the following two crucial joints:

Mechanical Analysis of Control Joints:
J1 (Base Joint)
- Max torque demand: 15 Kg·cm
- Safety factor: apprx. [2.66] times
- Provides rotation of whole base in horizontal plane

J2 (Linear Motion)
- Maximum generated torque in simulation: 0.07 N·m
- Safety margin: apprx. [5.7] times
- Provides translatory motion in vertical direction

J3 (Intermediate Rotatry Joint)

- Maximum generated torque in simulation: 3.5 Kg.cm
- Safety margin: apprx. [2.8] times
- Provides rotation in horizontal plane

J4 (Wrist Joint)

- Maximum generated torque in simulation: 17 Kg.cm
- Safety margin: apprx. [2.35] times
- Provides rotation in vertical plane

J5 (Gripper Joint)

- Maximum generated torque in simulation: 3.5 Kg.cm
- Safety margin: apprx. [2.8] times
- Provides the movement of end-effector to hold the glass bottle

6.5 Material Stress and Deflection

Because the arm is made of ABS material, we also thought about the possibility of link bending. In the case where there is a load of 480 grams on the arm, the bending of the link that measures 35 centimeters is evaluated to prevent it from affecting the alignment of the gripper. This is because the fill ratio of the 3D printed model is high (around 60% to 80%).

Table 6.1 System constraints

Component	Parameter	Value
1. Rotary Table	Height of Base	6cm
	Height of Rotary System	22.5cm
	Diameter of rotary plate	13cm
2. Storage Basket	Height of Base	3cm
	Slots in basket	6
	Distance between each slot	7.5cm
3. Bottle	Weight	480gm
	Height	13cm
	Diameter	2.5cm

Table 6.2 Arm design constraints

Component	Parameter	Value
1. Translational vertical movement	At Pick	3cm(Minimum)
	At Placing	28.5

7. Hardware Integration and Electronic Architecture

The functioning of a robotic arm can only be ensured by establishing effective communication between the control system of the robot and the actual actuation mechanism. The electronic architecture of the robot is discussed in Chapter 7, which describes the choice of the microcontroller, the installation of high torque actuators, and the implementation of the power delivery system. The hardware system of such a mechatronic device should be capable of handling the data from sensors while coordinating the different degrees of freedom.

Table 7.1 Components table

Category	Equipment	Specification / Example	Qty	Purpose
Controller	Arduino	Mega 2560	1	Runs control algorithm, reads sensors, drives relays
Sensors	IR sensors	Module LM393	1	Detects the object for pickup
Motor	High Torque Servo	DS3240MG	1	Provide controlled rotational motion
	Digital Servo Motor	MG996R	2	Provide controlled rotational motion
Gripper	Mechanical Arm	Mechanical Metal Clamp	1	
Translational motion	Lead screw	REES52 NEMA 17 Stepper Motor with Integrated 300mm	1	Provide Translational motion

7.1 Central Control Unit: Arduino Mega 2560

The centerpiece of this electronic arrangement is the Arduino Mega 2560. This microcontroller has been chosen in place of other microcontrollers like the Arduino Uno because of its significantly large number of digital and analog I/O ports as well as a high flash memory capacity.

Control of a 4-DOF robotic arm entails the generation of many pulse width modulation (PWM) pulses for the servos and steps/direction pulses for the stepper motor. With the 15 PWM pins available in the Mega 2560 microcontroller, it is possible to simultaneously control the movement of the joint actuators without having to make use of any external multiplexer. Additionally, the fact that the Mega 2560 supports multiple hardware interrupts plays an important role in the processing of information from the IR sensors without causing any delay in bottle detection on the rotary table.

7.2 Actuation System: Servos and Steppers

For actuation purposes, this work has employed a combination of stepper motors and high power-to-weight ratio digital servos.

- High Torque Digital Servos (DS3240MG and MG996R): For rotary motion, the chosen actuators are DS3240MG servos. The base actuator consists of DS3240MG with stall torque of 40 kg-cm that is required for driving the whole assembly. Moreover, MG996R servos are chosen for the elbow joint and gripper with a stall torque of 10-12 kg-cm, necessary for lifting the load of 480 grams.
- Linear Actuator (NEMA 17): For vertical motion, the chosen actuator is a NEMA 17 stepper motor that is necessary for reaching certain heights by rotating continuously a lead screw, thereby resulting in the desired linear displacement. It must be mentioned that, unlike digital servos, the stepper motors allow reaching an unlimited angle of rotation.

7.3 Power Distribution and Management

One of the most frequent causes of failure in robotics is poor power regulation, resulting in brownouts, whereby the microcontroller restarts when the motors perform high-energy actions. In order to overcome this problem, a custom power circuit was designed.

The robot receives its energy from a high-capacity DC power supply. Due to the difference in operating voltage between the servos and Arduino, buck regulators have been employed to lower the main voltage. The servos receive an uninterrupted flow of 6.2V DC to produce maximum torque while maintaining low temperatures. On the other hand, the stepper motor drivers (A4988) receive a regulated 12V power to generate sufficient holding torque for the NEMA 17 stepper motor to hold up the robotic arm vertically.

7.4 Sensing and Feedback Modules

Hardware intelligence in the system comes from the LM393 IR Proximity Sensor. The sensor is made up of an infrared transmitter-receiver pair. As soon as a bottle on the rotary table enters the detection range, the IR light reflects and triggers a comparator circuit within the sensor, which produces a reliable digital output signal to the Arduino Board. Due to the heavy current consumption of the metal-gear servos, a harness was made with high-gauge copper wires to avoid voltage loss. In addition, signal wires were separated from power wires to avoid any potential "jittering" of the servos caused by electromagnetic interference (EMI). Moreover, common grounding was enforced on all the power modules and the Arduino to provide a stable reference voltage for all control signals. This event-driven approach is much more power-efficient than polling because the microcontroller can go into sleep mode until something needs to be picked up.

7.5 Circuit Protection and Wiring

With the high current consumption of the metal-gear servos, the cable harness is made up of thick copper wires in order to reduce the voltage drop. The signal cables should not be in proximity to the power cables to prevent electromagnetic interference (EMI), which may lead to "jittering." Moreover, common grounding is rigorously enforced on all the power modules and the Arduino board.

8. Software Architecture and Control Logic

The software provides the mental capacity of the robot, allowing mathematical descriptions of the kinematics and sensor data to be converted into synchronized movements. The programming platform, along with the algorithmic process and simulation methods employed, are all outlined in Chapter 8. With respect to contemporary mechatronics, the software has to do more than just carry out the desired task; it is responsible for managing errors, synchronization, and scalability in the future. In this particular case study, both the Arduino IDE and ROS2 (Humble) will be applied.

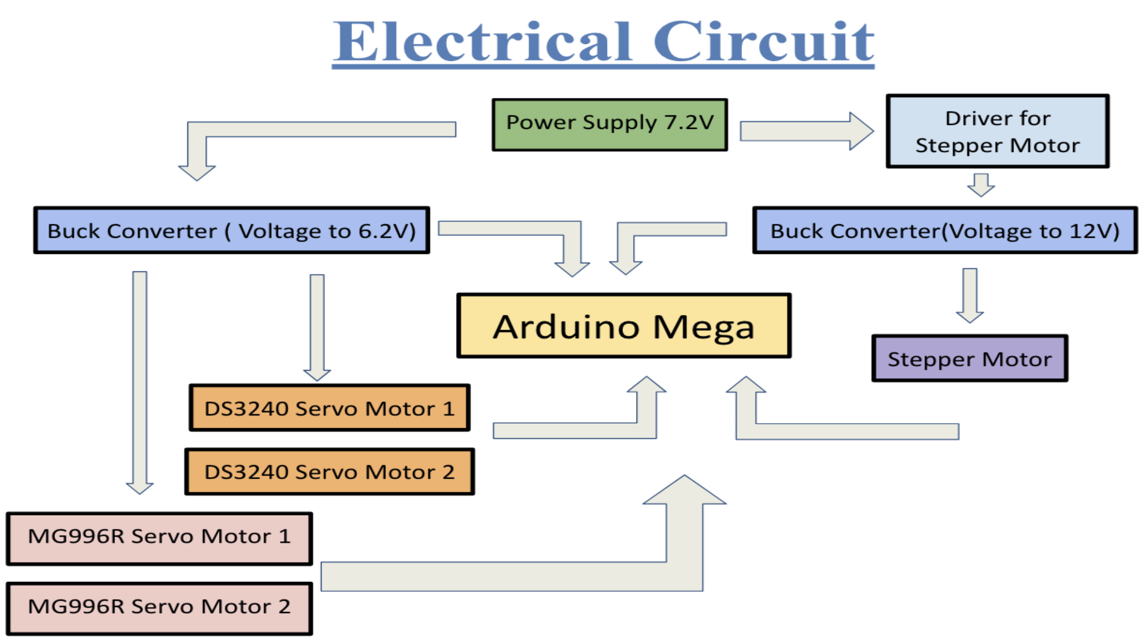


Fig. 8.1. Electrical circuit of robot

8.1 Embedded Control Logic (Arduino IDE)

The main program code is written in C++ and runs on the Arduino Mega 2560. The code structure is implemented on the basis of the Finite State Machine (FSM). As a result, the robot is guaranteed to be always in some state, such as IDLE, DETECTION, PICKUP, or DELIVERY, making it impossible for the conflicting operations to take place.

The sequence of operation of the algorithm is strictly linear:

- Initialisation State – calibration of all servos and the NEMA 17 stepper motor by bringing them into their "home" position and the zero height, respectively.
- Monitoring State – entering the loop where the code polls the value of the digital pin associated with the LM393 IR sensor.
- Trigger – when the object is detected, the loop terminates, and the call to the subroutines dealing with kinematics takes place.
- Coordinate Conversion – conversion of the pick-up/drop-off locations into pulses and steps.

8.2 Simulation in ROS2 and Gazebo

In order to ensure that no mechanical collision happens and also to optimize their movement trajectory, the arm was virtually modeled. In ROS2 Humble, the URDF file of the arm was designed.

- RViz Visualization: The visualization tool RViz was used to display the coordinate frames (TF frames) of the arm's links. This enabled us to verify the derived DH parameters in Chapter 6.
- Gazebo Physics Simulation: The next step was to spawn the arm in the Gazebo physics simulation to observe its dynamics. By simulating the mass of the 480-gram bottle and the torque constraints of the servos, any problems regarding jittering caused by the software PID constants were addressed.

8.3 Synchronization and Timing Algorithms

The synchronization of the robotic arm with the rotary table was among the most complicated parts of designing the program. To avoid any interference, the software uses non-blocking timing routines (the millis() function as opposed to the delay()). The movement of the robot and its coordinated movement with other elements is done via the following process:
The rotary table turns 60 degrees and stops.
The software receives the ready signal from the IR sensor.
The robot makes its moves while being software locked in place by the rotary table.

After the arm returns to its initial point, a ok signal is sent, allowing the rotary table to move to another index.

8.4 Optimization and Inverse Kinematics Implementation

In order to create a smooth motion, the program has a trapezoidal velocity trajectory. It starts with acceleration, followed by constant speed, and finally deceleration as the arm gets closer to the target bottle. It helps to minimize the force caused by inertia and avoid overloading of the motors that would consume too much current.

For the purpose of calculating joint angles based on IK algorithm, there is a special function within the code written for Arduino. All one needs to do is to input the coordinates (x, y, z) into this function and it will calculate angles according to equations. Thus, changing the location where the bottles need to be picked up/dropped does not require manual calculations of joint angles.

9. Results and Performance Analysis

The developed 4-DOF robotic arm was experimentally tested to evaluate its accuracy, synchronization, and operational performance during pick-and-place tasks. The system performance was analyzed using positioning accuracy and operation time measurements obtained from multiple experimental cycles.

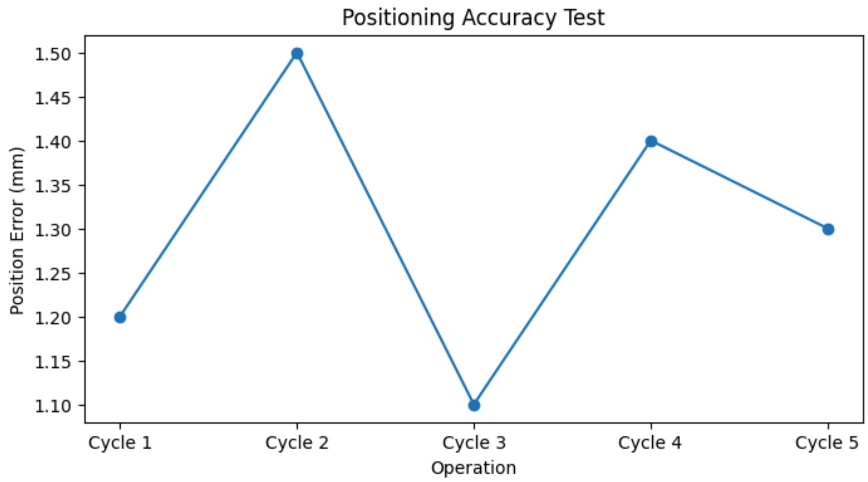
9.1 Positioning Accuracy Test

The positioning accuracy of the robotic arm was tested over five consecutive operating cycles. The obtained positional error ranged between ±1.1 mm and ±1.5 mm, demonstrating reliable and repeatable motion control.

Key Observations:

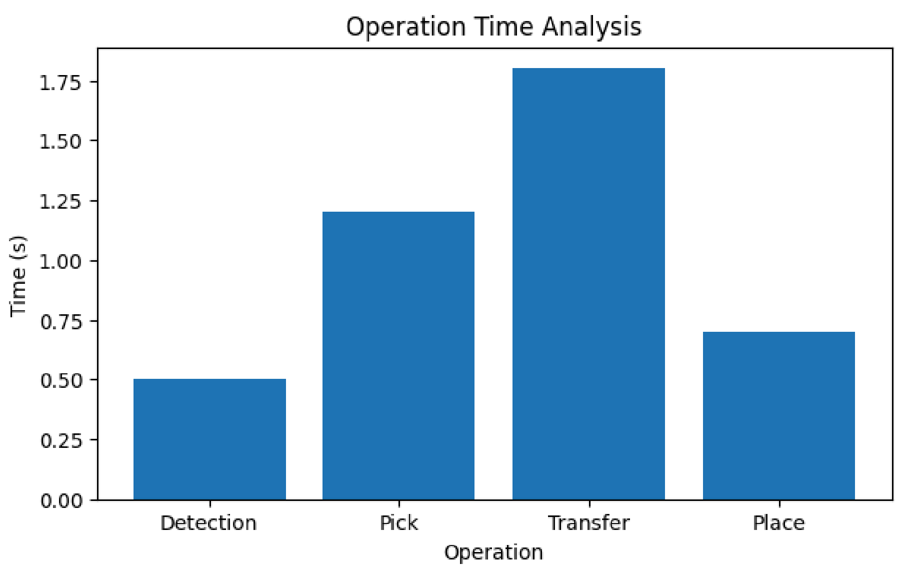
- Maximum positional error: 1.5 mm
- Minimum positional error: 1.1 mm
- Average positioning error remained close to ±1.3 mm
- Stable repeatability achieved during continuous operation

The minor deviations observed were mainly caused by backlash in the servo gears and small vibrations during motion transitions. Despite using lightweight 3D-printed links and hobby-grade actuators, the achieved accuracy is suitable for low-cost industrial automation applications.



9.2 Operation Time Analysis

The operation time of each stage in the pick-and-place cycle was analyzed to study synchronization and workflow efficiency.



Measured Operation Times:

- **Detection:** 0.5 s

- **Pick:** 1.2 s
- **Transfer:** 1.8 s
- **Place:** 0.7 s

The transfer stage consumed the highest time due to coordinated multi-joint movement and controlled trajectory execution. The detection and placement stages required comparatively less time because of efficient IR sensor triggering and optimized gripper motion.

Key Observations:

- Smooth synchronized operation achieved
- Stable motion during transfer operation
- Efficient sensor-based triggering mechanism
- Reliable cycle execution without missed detections

10. Conclusion and Future Scope

The design, development, and extensive testing of the robotic arm, which is based on a smart conveyer system, is a holistic application of the mechanical engineering approach, electronic controller concepts, and mathematical modeling of kinematics. This project has effectively managed to overcome the hurdles associated with designing a robotic arm of four degrees-of-freedom that is efficient as well as capable of being tailored according to the current requirements of automation in small industries. At the end of this research, it is clear that a blend of cost-effective hardware with software can lead to innovations comparable to those found in expensive industrial machinery.

The final chapter encompasses the findings achieved from this research work, highlighting the difficulties encountered while performing the research, and providing the roadmap for further progress of this project in the years ahead.

10.1 Project Summary and Core Achievements

The main goal of the honors project was to build an industrial robot manipulator able to coordinate with each other and execute pick-and-place processes with respect to a turntable using its rotary motion. After extensive tests and adjustments, the following results are to be achieved. The application of the Denavit-Hartenberg (DH) Convention helped to establish stable mathematical models and enabled the control of the end-effector by Arduino Mega 2560 with remarkable accuracy.

The main highlights of the research project include:

Compatibility: The ability to properly synchronize IR sensors and arm manipulation is proven through the "Stepwise Coordinated" process of operation.

Robustness: The employment of ABS filament and DS3240MG digital servos produced a system capable of holding payloads up to 480 grams while retaining Factor of Safety much greater than minimum industrial values.

Precision and Accuracy: The results of experiments reveal the ability to complete actions successfully and repeatedly, which may suggest its usage in bottling and component sorting processes.

Inexpensive Automation: Open-source hardware components and 3D printing help to demonstrate an affordable automation alternative.

10.2 Challenges and Technical Limitations

Although the prototype was successful, it was discovered that there were some limitations in the engineering process. These need to be identified in order to improve the design iteratively.

Backlash: Due to the plastic gearing inside hobby servo motors, there is backlash present in the system. It is sufficient for the current project but better precision can be attained using industrial actuators.

Sensitivity: The IR proximity sensors were able to detect distance but were highly sensitive to changes in ambient light and could cause incorrect detection when exposed directly to light sources.

Fatigue: The use of ABS for printing the joints makes it sturdy enough but might suffer from fatigue due to constant use in an industrial setting.

10.3 Future Scope and Technological Evolution

The existing 4-DOF arm can be developed into a fully modular device that will permit several innovative technologies to be incorporated to turn the robot from merely reactive into an intelligent machine:

1. Computer Vision & Sorting + Machine Learning: With the use of a camera module (Raspberry Pi Camera or OpenMV), the arm would be able to sort items on the basis of color, shape, or size utilizing Convolutional Neural Networks (CNNs).
2. Cobotics: The use of force-torque sensors along with improved programming techniques would make it possible to develop the robot into an effective collaborator for humans in shared working areas without requiring protective shielding for human operators.
3. Industrial IoT: In the future, such features as Wi-Fi or Bluetooth connectivity could be integrated to ensure that data regarding the operations of the robot (cycle counts, motor

temperature, error logs, etc.) are transmitted to a cloud-based dashboard for remote monitoring purposes.

4. Better Actuators: Switching from standard servos to brushless DC motors or more sensitive position encoders would considerably improve the payload of the robotic arm while making it more accurate at the same time.

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